Experiment Set 2

Submission Due Date: 18/09/2026

Experiment 11: Review of Components

With help of Multi-meter, perform following experiments.

Task (11.a) Measure values of different resistors.

Task (11.b) Take a variable resistor (Potentiometer). Measure the values of resistance with help of multi-meter for three different positions of the potentiometer settings.

Task (11.c) Measure at least five value of LDR for various light intensities. Observe how resistance changes with light intensity.

Answer Following Questions in your report..

Question 11.a: For the Task (11.a), take three photos of measurements of three different resistances and include them in your report.

Question 11.b: For the Task (11.b), take three photos of measurements of three different resistances based on three different settings of the variable resistance and include them in your report.

Question 11.c: For the Task (11.c), take five photos of measurements and draw an approximate graph (X-axis: Light Intensity; Y-axis: LDR resistance) to represent your observations. Explain why LDR resistance changes with the change in the light intensity.

Answer Following Questions in your report..

Answer the following with reference to the LDR Datasheet.

Question 11.d: Which material the LDR is made of?

Question 11.e: What is power dissipation rating at 30°C ?

Question 11.f: What is the operating Temperature Range?

Question 11.g: What is the minimum Dark Resistance?

Question 11.h: At which wavelength, the relative spectral response if the highest?

Experiment 12: Building Potential Divider Circuit

Task (12.a): Connect five resistors in series. Connect one end to +5 V from Arduino and other end to Ground Pin of Arduino.

Task (12.b): Measure voltage values on different combinations of Resistors.

Answer Following Questions in your report..

Question 12.a: For the Task (12.a), take one photos showing the connections and include it in your report.

Question 12.b: For the Task (12.b), take three photos of measurements of three different combinations of resistances. Each photo should clearly show the Voltage measured. Include these photos in your report.

Question 12.c: Write the mathematical equation that determines the voltage across one resistor in the series with the total five resistors.

Experiment 13

Task (13.a): Connect LDR in series with one Resistor. Connect one end to +5 V from Arduino and other end to Ground Pin of Arduino.

Task (13.b): Measure voltage across LDR for three different Light intensities on LDR.

Answer Following Questions in your report..

Question 13.a: For the Task (13.a), take one photos showing the connections and include it in your report.

Question 13.b: For the Task (13.b), take three photos of measurements of voltage across the LDR with three different Light Intensities on it. Each photo should clearly show the Voltage measured. Include these photos in your report.

Question 13.c: Write the mathematical equation that determines the voltage across LDR in the series with the resistor.

Experiment 14

1. Use Program Titled “Experiment_14”
2. Connect circuit as mentioned in the comment of above program. Read the comments in IDE window.
3. The aim of the program is to observe how blinking rate of the LED changes with change in position of potentiometer.
4. Execute the program and observe the output on the hardware. Observe how the blinking rate of LED changes with the change in the position of potentiometer.

Understand the complete program written on next page.

```
int sensorPin = A0;    // select the input pin for the potentiometer
int ledPin = 13;      // select the pin for the LED
int sensorValue = 0;  // variable to store the value coming from the sensor

void setup() {
  // declare the ledPin as an OUTPUT:
  pinMode(ledPin, OUTPUT);
}

void loop() {
  // read the value from the sensor:
  sensorValue = analogRead(sensorPin);
  // turn the ledPin on
  digitalWrite(ledPin, HIGH);
  // stop the program for <sensorValue> milliseconds:
  delay(sensorValue);
  // turn the ledPin off:
  digitalWrite(ledPin, LOW);
  // stop the program for for <sensorValue> milliseconds:
  delay(sensorValue);
}
```

Answer Following Questions in your report..

Question 14.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 14.b Explain the following function.

- ✓ analogRead

Question 14.c: Draw the schematic diagram that represents the internal structure of potentiometer. From this diagram, explain how we can get variable values of resistance.

Answer Following Questions in your report..

Question 14.d: Find the diagram/photo of ATmega328P 28-Pin DIP configuration. In this diagram/photo, clearly designate and illustrate the mapping of the following pins of Arduino UNO with corresponding Port Pins of ATmega328P 28-Pin DIP .

(14.d.1) All six Analog input pins

(14.d.2) All Digital Input / Output Pins

(14.d.3) All the Pins with PWM facilities

Answer Following Questions in your report..

Question 14.e: For all the six Analog input pins of the Arduino UNO as described in (14.d.1), explain which feature of the microcontroller enables the analog input.

Question 14.f: For all the Pins with PWM facilities, describe how the microcontroller enables the PWM mode of operation.

Experiment 15

1. Keep the hardware similar to Experiment 14.

2. Change the program such that in place of blinking rate of LED, the intensity of LED changes with the position of the potentiometer.

Save the new program.

Answer Following Questions in your report..

Question 15.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 15.b: Include your new c program in your report.

Question 15.c: Describe the changes you have made in the program.

Experiment 16

1. Modify the input side of the circuit of the Experiment 14. In place of potentiometer, use LDR.

2. Learn how to connect LDR to input pin of Arduino. Discuss your learning and the input circuit with the instructor.

3. Use the same program of Experiment 14.

4. Observe how the blinking rate of LED changes with the change in light intensity on LDR.

Answer Following Questions in your report..

Question 16.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Experiment 17

1. Keep the hardware similar to Experiment 16.

2. Change the program such that in place of blinking rate of LED, the intensity of LED changes with the light intensity on the LDR.

Answer Following Questions in your report..

Question 17.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 17.b: Include your new c program in your report.

Question 17.c: Describe the changes you have made in the program.

Ethernet Shield Experiments

Experiment_18

1. Use Program Titled “Experiment_18”
2. Connect appropriate sensors to analog input pins described in the program.
3. Observe the output/display after successfully executing the program.

Answer Following Questions in your report..

Question 18.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 18.b: Study the program and answer following questions.

(18.b.1) Which library files are included ?

(18.b.2) How “mac address” is specified ?

(18.b.3) How “IP address” is specified ?

(18.b.4) How data are transferred on internet ?

Answer Following Questions in your report..

Question 18.c: When you connect Ethernet Shield with Arduino UNO, which pins of Arduino are utilized in connecting it to internet? Use a diagram/photo of ATmega328P 28-Pin DIP configuration. In this diagram/photo, clearly designate and illustrate the mapping of the pins of Arduino UNO with corresponding Port Pins of ATmega328P 28-Pin DIP. Describe each pin involved with their functions. Explain which facilities/features of these microcontroller pins make the internet connection possible.

Answer Following Questions in your report..

Question 18.d: Explain following functions in detail.

(18.d.1) Serial.begin(9600);

(18.d.2) Ethernet.begin(mac, ip);

(18.d.3) Ethernet.hardwareStatus()

(18.d.4) Ethernet.linkStatus()

(18.d.5) server.begin();

(18.d.6) EthernetServer server(80);

(18.d.7) Serial.println ();

(18.d.8) Serial.print ();

(18.d.9) client.println();

(18.d.10) client.print();

Experiment 19

1. Begin with the program of Experiment 18.
2. Modify above program in following way. Display the values on only two analog input pins A1 and A4 as described below.
   a) Connect LDR to A1 input and display its value with text “The current LDR value is .. ”
   b) Connect potentiometer to A4 input and display its value with text “The current Potentiometer value is .. ”
3. Change above inputs (light intensity on LDR and position of potentiometer) and observe the change in the o/p on the web address.

Answer Following Questions in your report..

Question 19.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 19.b: Include your new c program in your report.

Question 19.c: Describe the changes you have made in the program.

Experiment 20

1. Begin with the program of Experiment 19.

2. Modify above program in the following ways. Connect potentiometer to A0 input and display its value with the text on the “Serial Monitor” as well as on the website “ThingSpeak.com”

Answer Following Questions in your report..

Question 20.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 20.b: Include your new c program in your report.

Question 20.c: Describe the changes you have made in the program. Write down the differences in programs for experiments 19 and 20.

Answer Following Questions in your report..

Question 20.d: Write the steps how will you open an account on ThingSpeak.com.

Question 20.e: Describe the details in steps about how will you send the data from A0 pin of the Arduino UNO to this website.

Question 20.f: Describe the details of how you are sending the data to the Serial Monitor.